

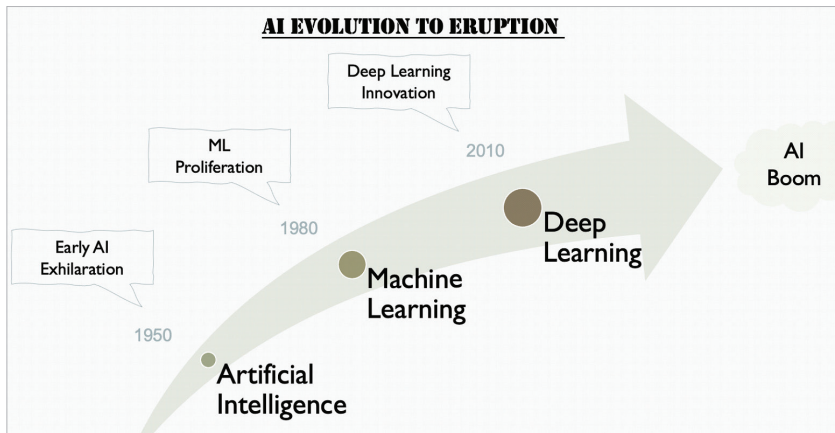


Figure 2: Evolution of AI, ML and DL

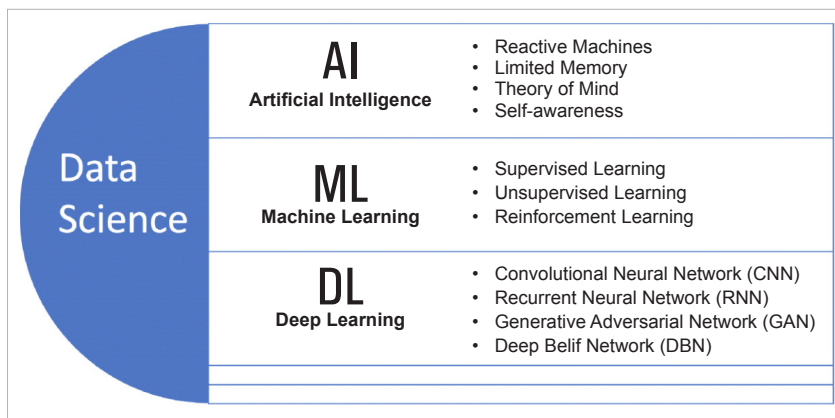


Figure 3: Types of AI, ML and DL

created the need for data analytics, more popularly known as data science, leading to the evolution of machine learning as an approach to achieving artificial intelligence.

Machine learning

ML is the use of algorithms to process, learn and make sense or predict the pattern of available data. More recently, the low-code and no-code concepts of software development are being used in machine learning as self-learning processes that give specific instructions to accomplish particular tasks. The machine is 'trained' by using data and algorithms, giving it the ability to learn how to perform the task and, more importantly, apply the learning to evolve continuously.

ML was evolved when the developer community focused on AI, and then developed algorithmic decision-tree learning, logic programming, clustering, parallel processing and reinforcement learning.

These were all good steps in the right direction but not enough to solve use cases that were of interest to the world.

Deep learning

DL is an evolution of neural networks and machine learning, and the brainchild of the AI community. It learns about how the human mind works in specific scenarios, and then gets better at that job than humans! As an example, IBM's Watson played chess against itself and improved at the game so much to eventually beat the world champion. Google's AlphaGo also learnt how to play the Go board game by playing it over and over to better itself, and became the champion.

AI, ML and DL are evolving continuously. It's the intent of everyone involved with data science to advance these concepts to better our daily lives. The good thing is that the open source community, private enterprises, scientists, and government agencies are all working together for this.

To conclude, while AI helps to create smart intelligent machines, ML helps to build AI-driven applications. DL is a subset of ML; it trains a specific model by leveraging complex algorithms for large volumes of data. As narrow AI is extremely difficult to develop, ML is addressing the opportunities in this space with rigid computing. DL helps to bring AI and ML together, at least for realising general AI. **END** 🐧

References

- [1] The History of Artificial Intelligence: <https://sitn.hms.harvard.edu/flash/2017/history-artificial-intelligence/>
- [2] Machine Learning Group at MIT: <https://machinelearning.mit.edu/>
- [3] Deep Learning at IBM: <https://www.ibm.com/cloud/learn/deep-learning>

By: Bala Kalavala

The author is a seasoned emerging technical architect, evangelist, thought leader and a sought-after keynote speaker. He currently works as a distinguished member of the technical staff and heads the Enterprise Architecture practice as chief architect at Wipro Ltd.

Disclaimer: The views expressed in this article are that of the author and not of the organisation he works in.